

Boyce di prima / Chap. 7 ; Sistemas EDO 1^{er} orden / Chap 10 ; EDP / Chap 11 ; Prob. Sturm Liouville

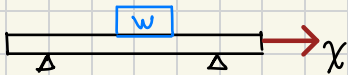
Eq de movimiento $m \frac{d^2 U}{dt^2} = -kU$ Termicos $m \cdot C_p \frac{dT}{dt} = h(T_{\infty} - T)$

Discretos $\rightarrow$ Vector de estados escrito con n valores finitos
 $\hookrightarrow$ EDO

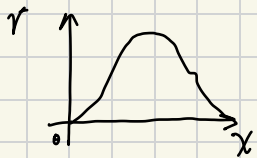
$$\begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} (t) \quad n < \infty$$

Continuos (Ecuaciones Diferenciales Parciales) $\rightarrow$ Simplificación

1D
2D
3D

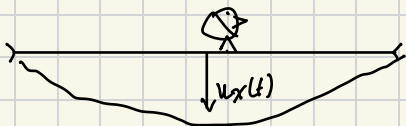
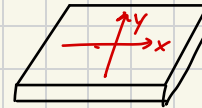
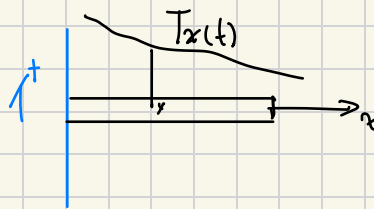


Variable de estado $\rightarrow$ deflexión de la viga.

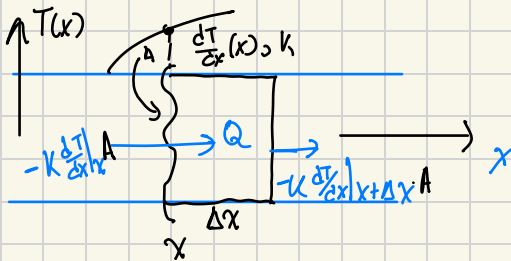


$u(x,t)$
 $u_x(t) \sim u(t)$

no se puede partir en n masas.



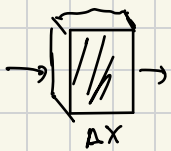
Problema difusivo Continuo



Fourier Ley Constitutiva, depende del material

$$Q = -K \frac{dT}{dx} A$$

Pasamos de estas temperaturas a estas derivadas

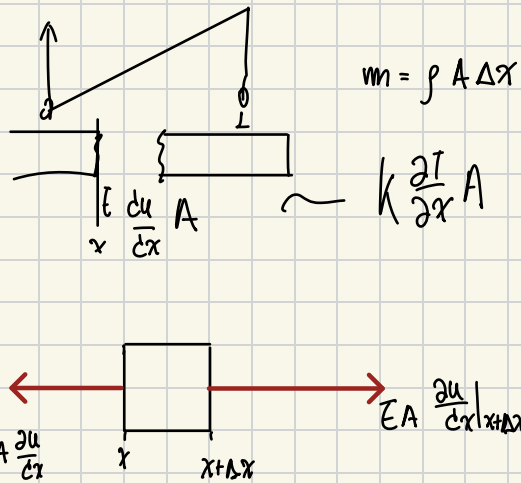
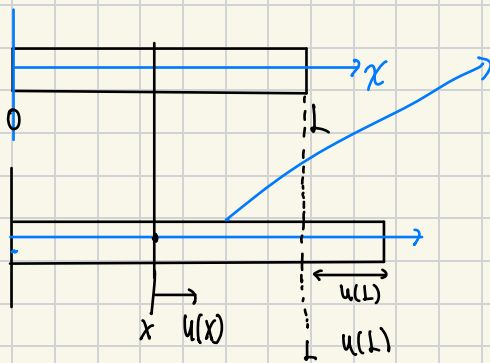


$$\underbrace{-k \frac{dT}{dx} \Big|_x A}_{\text{Entra}} + \underbrace{k \frac{dT}{dx} \Big|_{x+\Delta x} A}_{\text{Sale}} = \rho A \Delta x C_p \frac{dT(t)}{dt} \rightarrow \rho A C_p \frac{\partial T(t)}{\partial t} = \lim_{\Delta x \rightarrow 0} \frac{K \left(\frac{dT}{dx} \Big|_{x+\Delta x} - \frac{dT}{dx} \Big|_x \right) A}{\Delta x}$$

$$\frac{du}{dt} = \sum Q$$

$$\rho C_p \frac{\partial T(t)}{\partial t} = K \frac{\partial^2 T}{\partial x^2} \quad \begin{array}{l} \text{Eq Calor /} \\ \text{Eq de difusión} \end{array}$$

Problema mecánico Continuo



$$\rightarrow \rho A \Delta x \frac{\partial^2 u}{\partial t^2} = E \frac{\partial^2 u}{\partial x^2}$$

Discreto

Continuo 1-D

$$[M^T] \frac{d\vec{T}}{dt} = [K] \vec{T} + \vec{q} \quad \text{Ley}$$

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

$$[M] \frac{\partial^2 \vec{u}}{\partial t^2} = [K] \vec{u} + \vec{f} \quad \text{Newton}$$

$$\rho \frac{\partial^2 u}{\partial t^2} = E \frac{\partial^2 u}{\partial x^2}$$

Estado

$$\begin{pmatrix} T_1 \\ \vdots \\ T_n \end{pmatrix} (t) \quad \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} (t)$$

Estado

$$\begin{matrix} T(x,t) & u(x,t) \\ T_x(t) & u_x(t) \end{matrix}$$